Validation of a Multiscale In Silico Framework for Coronary Atherosclerosis Progression in a Porcine Model

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Introduction

Atherosclerosis is a multifactorial vascular pathology driven by intricate interactions between systemic risk factors, localized hemodynamics, and dynamic cellular responses. Endothelial permeability under disturbed blood flow enhances subendothelial low-density lipoprotein (LDL) accumulation, triggering leukocyte infiltration, cytokine release, and vascular smooth muscle cell (vSMC) phenotypic transition. Although discrete biological models and fluid dynamic simulations have addressed these phenomena individually, integrating multi-physics processes into a cohesive predictive pipeline remains challenging. In this study, we present the validation of a hybrid multiscale *in silico* framework designed to forecast long-term coronary plaque initiation, morphology, and cellular composition against longitudinal *in vivo* porcine data.

Methods

The computational workflow couples three distinct spatial and temporal scales: (1) 3D transient computational fluid dynamics (CFD) based on patient-specific reconstructed geometries to compute time-averaged wall shear stress (TAWSS); (2) a 2D mass transport model solving convection-diffusion-reaction equations to simulate oxLDL accumulation within the intima; and (3) a discrete agent-based model (ABM) regulating cellular events, including macrophage activation, foam cell conversion, and vSMC recruitment and phenotypic modulation..

Validation was conducted using longitudinal imaging data from the BIOCCORA study, involving four coronary arteries (two LAD and two RCA) from hypercholesterolemic pigs monitored via IVUS and OCT at baseline (3 months) and 9-month follow-up. Model performance was evaluated across 50 cross-sections per vessel. Primary validation metrics included quantitative lumen stenosis ratio (SR) and local wall thickness change, accompanied by non-parametric Wilcoxon signed-rank tests to evaluate paired bias.

Results

The multiscale model successfully reproduced the spatial growth patterns across different hemodynamic environments. Regions exposed to high TAWSS exhibited minimal to no remodeling, confirming the atheroprotective role of elevated shear stress. Conversely, low-TAWSS sectors developed pronounced eccentric thickening, lipid-rich necrotic cores, and adaptive vessel expansion consistent with Glagovian remodelling. Quantitatively, the framework demonstrated high predictive accuracy for wall thickness variation. Stenosis ratio (SR) predictions closely tracked spatial trends with minor local discrepancies occurring predominantly in highly eccentric, lipid-dense lesions

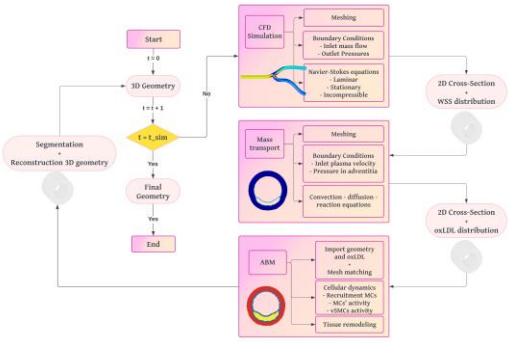


Figure 1: Overview of the multiscale computational pipeline integrating CFD, mass transport, and ABM for plaque progression.

Conclusions

This study demonstrates the capability and robustness of a hybrid CFD-transport-ABM framework to quantitatively predict regional coronary plaque progression and tissue composition without reliance on post-hoc empirical curve fitting. By capturing emergent mechanobiological interactions, this approach provides a solid foundation for preclinical risk stratification, target identification, and patient-specific *in silico* clinical trials..

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